

Convertidor CD-CD 12v - 127 V, 100 W , frecuencia de 100 kHz

10 % de rizado de voltaje de salida y 25 % de rizado en la corriente del inductor.

Boost

$$f = 100 \text{ kHz}$$

$$P = 100 \text{ W}$$

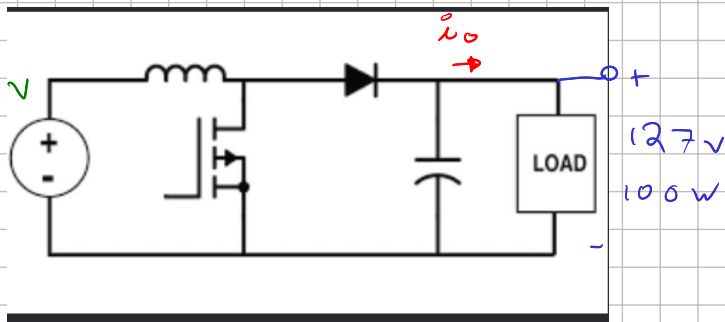
$$V_o = V_{in} \left(\frac{1}{1-d} \right)$$

$$L = \frac{V_{in} d}{f \cdot \Delta i}$$

$$i_{in} = \frac{P_{out}}{V_{in}} = \frac{100}{12}$$

$$C = \frac{i_o \cdot d}{\Delta V_o \cdot f}$$

$$i_{in} = 8.33 \text{ A}$$



$$i_o = \frac{P_o}{V_o} = \frac{100 \text{ W}}{127 \text{ V}} = 787 \text{ mA} \approx 0.79 \text{ A} \quad (2)$$

$$\Delta i_L = 25\% i_i = 25\% \cdot 8.33 \text{ A}$$

$$\Delta i_L = 2.0825 \text{ A} \quad (3)$$

$$d = 1 - \frac{V_{in}}{V_o} = 0.9 \quad (1)$$

$$\Delta V_o = 10\% V_o = 12.7 \approx 13 \text{ V} \quad (4)$$

$$L = \frac{12 (0.9)}{100 \text{ k} (2.0825)} = 51.86 \mu\text{H}$$

$$C = 560 \text{ nF}$$